Multi-Factor ETF Strategy with Automated Risk Management

Technical Investment Document — *Living Edition*

Quantitative Portfolio Management
Repository: `stuarrob/code` (branch `main`)

Last major review: 2026-07-09
Continuously maintained; see §?? and Change Log

Abstract

This is the **operating document** for a live-money systematic ETF strategy running against an Interactive Brokers account. It is designed to be read *instead of the code*, and to remain accurate to the code as the code evolves. It combines a specification of the strategy, a canonical statement of the currently-live policy values, an adversarial review of known limitations, and a change log.

The strategy selects ETFs from a broad US-listed universe using a weighted geometric mean of four factors: momentum, quality, low-volatility, and value. Factor weights (currently 35/30/20/15) are held as an *informed prior* rather than an in-sample optimum, on ergodicity-economics grounds described in §??. Portfolio construction is rank-based with top-N selection (currently N=20) and exponential rank weights. Risk management is via percentage-based trailing stops placed on every entry.

The document is deliberately organised so that (a) values which can drift are pulled into a single *Canonical Live Configuration* section (§??) that mirrors `configs/etf_smart_beta.toml`, (b) the strategy narrative in the body describes the design intent regardless of specific parameter values, and (c) an *Adversarial Review* section (§??) records known flaws with dated action plans. Historical backtest metrics referenced in the body describe design-time analyses; live-account performance is reported separately in the change log.

Reader assumption: you are a serious investor with real money in this system, not a marketing target. Numbers that are unknown or degraded are stated as such rather than papered over.

Contents

1 Living Document Compact

1.1 Purpose and Reader Contract

This document is the single reading surface for the strategy. The operator has explicitly declared that they read this document rather than the code as the primary way to understand what the system does and why. That places three obligations on the document:

1. **Accuracy over marketing.** If the live configuration differs from a value described in a body section, the discrepancy is a documentation bug and must be reconciled. Never present a stale spec as if it were current.
2. **Honesty over optimism.** Known limitations get their own section (§??) and remain there until fixed. Backtest performance is separated from live performance; both are labelled.
3. **Provenance.** Every material claim traces to either (a) the code / policy file (with path), (b) a dated research note under `docs/research/`, or (c) published academic literature. If a claim traces only to intuition, it says so.

1.2 How to Keep This Document Living

The canonical sources of truth are, in order of authority:

1. `configs/etf_smart_beta.toml` — the live policy parameters (weights, position count, bounds, stops, etc.).
2. Files under `src/` — the deterministic pipeline. In case of conflict with the doc, code wins and the doc gets fixed.
3. `docs/research/` — dated research notes justifying policy values.
4. `docs/RESEARCH_BACKLOG.md` — open questions and their status.
5. `git log` — authoritative for "when did this change and why."

Update triggers. This document must be edited (and the Change Log at §?? appended) whenever:

- A value in `configs/etf_smart_beta.toml` is changed.
- A factor definition changes in `src/factors/`.
- A risk-management rule changes in `src/portfolio/risk_manager.py` or the execution scripts.
- A new research note lands under `docs/research/`.
- An entry in the Adversarial Review (§??) is closed out, opened, or reprioritised.
- Real-money performance is measured at any milestone (quarter-end, year-end, or notable drawdown/rebound).

If a change is made to code or policy without a corresponding doc edit, that is a bug against *this document*, not the code.

1.3 Known Drift Between Body Sections and Live Configuration

The body of this document (starting at §??, "Changes from Version 3.0 and Why They Matter", through the Factor Framework and Portfolio Construction sections) describes a strategy variant that was designed and backtested in early 2026 — 30 positions, tight 3–8% weight bounds, robust MVO as default, quarterly rebalance. That variant is a legitimate design that produced the backtest metrics referenced.

The live policy at the time of the last major review (2026-07-09), captured in §??, does not match that variant in three places: position count (20 vs 30), weight bounds (2–15% vs

3–8%), and default optimiser (RankBased in scripts vs MVO in the design). This drift is tracked as item P1 in `docs/RESEARCH_BACKLOG.md`. Until resolved, **treat the Canonical Live Configuration section as authoritative for what is actually running**, and the design sections as historical context.

1.4 Design Principles

Three principles govern every design choice in this system. They are non-negotiable: proposals that violate them are rejected regardless of in-sample Sharpe. New features must be evaluated against each before implementation.

Principle 1 — Deterministic decides, LLM narrates

All numbers that reach the broker (selections, weights, order sizes, stop levels) come from deterministic code in `src/`. Any LLM in the applet's Explain step only narrates the deterministic output. Under no design does a language model modulate a size, a weight, or a target position. Auditability requires that every payload be traceable to a pure function of policy and market data. Discussed further in the `ADR-0001` portfolio-applet decision record.

Principle 2 — No magical thinking on the knobs

Every numeric parameter in `configs/etf_smart_beta.toml` must trace to either published research or a dated backtest recorded in `docs/research/`. If the trace ends at "seemed reasonable," it belongs on `docs/RESEARCH_BACKLOG.md` with a target for empirical justification. Semi-annual revisits of the calibration are scheduled. This rules out both unaudited human intuition *and* silent numerical drift from copied defaults.

Principle 3 — Slowly varying only; whipsaw is a disaster

Whipsaw — any part of the system flipping decisions on noise — is treated as evidence of overfitting, not evidence of alpha. In a non-ergodic system (§**??**), whipsaw compounds negatively in log-space: each unnecessary round-trip erodes the time-average growth rate even when the arithmetic-mean signal looks unchanged. Concrete implications for every layer:

- **Long lookbacks are preferred over short.** The 252-day momentum and 60-day volatility windows are conservative choices that smooth noise. Any proposal to shorten a lookback must include an explicit whipsaw analysis on historical data.
- **Rebalance drift thresholds err high.** The current 5% threshold produces ≈6 rebalances per year. Tightening it to 2–3% would produce more trading on smaller deviations — textbook whipsaw. Wider thresholds are considered before tighter ones.
- **Regime overlays require hysteresis.** If a market-regime signal is added (e.g. SPY vs 200-day moving average with a VIX guard, per adversarial-review item T2.1), it must include (a) N-day smoothing on the trigger and (b) a minimum dwell time of ∼30 days in either regime state before switching back. Flicker weekly and the overlay is worse than not having it.
- **Factor timing is rejected on principle.** No design that dials individual factor weights up or down based on recent factor performance — even if it looks like it would have worked in-sample. Factor timing is the archetypal whipsaw trap; the ergodicity argument amplifies its cost.
- **Trailing stops must be calibrated to avoid market-noise triggers.** The current 10% trailing width sits above typical intraday and one-week volatility for the universe. Anything materially tighter would trigger on ordinary noise.

- **Correlation-clustering (when implemented, T2.2) uses a long window and updates infrequently.** Clusters computed on 6-month pairwise correlations and refreshed quarterly — not weekly.
- **Sentiment overrides at most semi-annually.** Discretionary weight tilts by the operator (see §**??**, "When to revise") are a rare, deliberate act — never a reaction to weekly headlines.
- **Fundamental-data revisions (once value factor lands, T1.1) get smoothing.** Restated earnings and updated P/E ratios do not immediately propagate; changes are smoothed over ≥30 days.
- **Any signal that fires and reverses within one rebalance period is noise.** If a proposed feature would flip a position in and out on a monthly cadence in backtests, it is rejected regardless of headline Sharpe.

The three principles together create a filter that admits deliberate, evidence-anchored, slowly-varying changes and refuses reactive ones.

2 Canonical Live Configuration

The values below are the current live policy. They mirror `configs/etf_smart_beta.toml`. When the TOML changes, this section changes and the Change Log records the diff.

2.1 Portfolio Construction Parameters

Parameter	Value	Justification
`num_positions`	20	Historic default; drift from tech-doc target of 30 tracked in backlog
`min_weight`	2%	Historic default
`max_weight`	15%	Historic default; tech-doc target 8%
`tax_status`	`taxable`	Operator declared taxable status

2.2 Factor Weights (informed prior)

Factor	Weight	Justification
momentum	35%	See §**??** — informed prior
quality	30%	Insurance premium against drawdown asymmetry
volatility	20%	Insurance premium; low-vol anomaly
value	15%	**Proxied by expense ratio only — see §?? T1.1**

2.3 Factor Calculation Windows

Parameter	Value
`momentum` lookback	252 trading days (AQR / Jegadeesh-Titman)
`momentum_skip_recent`	21 trading days (Novy-Marx 2012 short-term reversal skip)
`quality` lookback	252 trading days
`value` lookback	252 trading days
`volatility` lookback	60 trading days

2.4 Rebalancing

Parameter	Value	Justification
`frequency`	`bimonthly` (~6/yr)	Historic default; tech-doc target quarterly
`drift_threshold`	5%	Round number; not calibrated

2.5 Risk Management

Parameter	Value	Justification
`entry_stop_loss_pct`	12%	Round number; not calibrated
`trailing_stop_pct`	10%	Round number; not calibrated
`cash_reserve`	$70,000	Round number; ≈18% of current NAV

2.6 Optimiser Priors (used by MVO variant)

Parameter	Value	Notes
`risk_aversion`	1.5	Standard MVO λ; risk-tolerant end of retail range
`robustness_penalty`	0.7	Axioma γ; aggressive; not calibrated
`turnover_penalty`	0.2	Medium; not calibrated

3 Adversarial Review and Known Limitations

This section records material flaws in the strategy in the operator's own commissioned adversarial review (2026-07-09). Items remain here until closed; closed items move to the Change Log with the date and commit that closed them. Tiering is by impact-on-live-money:

- **Tier 1 (T1)** — material design flaw; can cause the strategy to underperform or misbehave in ways the operator does not consent to.
- **Tier 2 (T2)** — significant weakness; should be addressed before scaling capital or before a live BIG-switch (§**??**).
- **Tier 3 (T3)** — smaller nudge; worth investigating; unlikely to change conclusions materially.

3.1 T1 — Material design flaws

3.1.1 T1.1 The value factor is fake

`src.factors.SimplifiedValueFactor` returns $-1\times$expense ratio. This is a low-fee preference, not a value factor. Real value factors use fundamental multiples (P/E, P/B, dividend yield, EBITDA/EV). Consequences:

1. The integrated score is genuinely a three-factor score, not four. The 15% weight allocated to "value" is buying almost nothing.
2. The natural regime hedge that value provides — value tends to outperform when growth/-momentum breaks (2000–2002, 2022) — is absent from the model.
3. In an AI-tech unwind scenario, this is the single largest reason the strategy has less protection than the factor weights suggest.

Fix. Replace the expense-ratio proxy with a fundamental composite (P/E, P/B, dividend yield) pulled from a data source. For ETFs, per-ticker fundamentals aggregate to weighted-average holding metrics; `yfinance ticker.info` exposes several. Estimated effort: 1–2 evenings.

Priority: highest single-value fix in the system.

3.1.2 T1.2 Survivorship bias in the cached universe

The per-ticker parquet cache at `~/trade_data/ETFTrader/ib_historical/` contains only ETFs that still trade. ETFs that were delisted, merged, or liquidated over 2021–2026 are absent. Every backtest run on this cache overstates historical performance by the average return of the survivors relative to the true universe. The effect is smaller for ETFs than single stocks (ETF closures are rarer, mostly small AUM), but non-zero.

Fix. True fix (point-in-time universe reconstruction) requires Bloomberg / CRSP-tier data and is out of scope. **Realistic mitigation:** apply a 1–2% haircut to backtested CAGR whenever it is quoted, and cross-check against SPY over the same window. Do not treat backtested CAGR as a forward expectation.

3.1.3 T1.3 Look-ahead bias is presumed absent but never proved

Factor calculations are called with `prices.loc[:rebalance_date]` in the diagnostic pipeline. This should mean no factor sees future data. But no pinning test exists that asserts this. If any factor calculation quietly uses `.mean()` or `.std()` over a window that includes $t+1$ data, the backtest is silently inflated.

Fix. Add a test to `tests/test_factors/` that: (a) computes factor scores at date t on a small synthetic price series, (b) shifts prices by 1 day and recomputes, (c) asserts the score at t is unchanged. Estimated effort: 2 hours. **Should have been part of the initial factor test suite.**

3.2 T2 — Significant weaknesses

3.2.1 T2.1 No regime awareness

The strategy runs identically in bull, bear, and crisis regimes. Long-only, concentrated (top-20), 22% cash reserve, 10% trailing stops. There is no signal to de-risk in a rising-VIX / breadth-collapsing environment. The 22% cash reserve is a *fixed allocation*, not a regime-responsive hedge.

Fix. Add a portfolio-level trend filter as an overlay. Rule: if SPY closes below its 200-day moving average AND VIX > 25, shift target equity exposure from 100% to 60%, hold the remainder in short-term Treasuries (SHV or BIL). This is documented crash protection — not market-timing — and has substantial backtest support in the Meb Faber IVY portfolio literature. Estimated effort: 1 evening.

3.2.2 T2.2 Correlation blindness in top-N selection

The integrated factor score picks the top-N ETFs. It does not check whether the top-N are highly correlated with each other. The current 28-position live holdings read as a broad international dividend / value / quality tilt with likely pairwise correlations of 0.6–0.8 across most positions. The effective portfolio Sharpe is lower than the individual position Sharpes would suggest.

Fix. In the selection step: rank candidates by score, cluster the top 2N by pairwise return correlation (hierarchical clustering over the last 6 months), then pick the top-scored representative from each cluster until N positions are filled. Same signal, better diversification. Estimated effort: 1 evening.

3.2.3 T2.3 Conviction magnitude is discarded

The integrated factor score has a magnitude that is not used. Position #1 at rank 0.95 and position #20 at rank 0.62 are subject to the same weight bounds. The RankBased exponential weighting partially mitigates but is coarse. The MVO variant uses score-as-alpha but with tight bounds that suppress conviction.

Fix. Feed score magnitude into the optimiser as expected-return alpha, allowing tighter clustering around the strongest signals. Estimated effort: 1 evening. Lower priority than T2.1 and T2.2.

3.3 T3 — Smaller nudges

3.3.1 T3.1 Factor lookback windows are convention, not calibrated

252/21/60 are the AQR convention (Jegadeesh-Titman 12-1; standard 60d vol). Never tested for optimality in this universe. Momentum is relatively robust to lookback choice; volatility less so. A grid search over lookback pairs would be cheap and informative. Estimated effort: 1 evening.

3.3.2 T3.2 Risk parameters are round numbers

Trailing 10%, entry 12%, cash $70k, drift 5%, weight bounds 2–15%. All defensible, all round, none calibrated. A sensitivity grid would locate them on the risk-return frontier. Estimated effort: 1 weekend.

3.3.3 T3.3 Rebalance frequency untested

Bimonthly (6/yr) vs quarterly (4/yr). Turnover cost vs signal freshness trade-off never explicitly measured on this universe. Estimated effort: 1 evening.

3.4 Prioritised Fix Order

Recommended sequence to address the flaws above:

1. **T1.3** (look-ahead pinning test) — 2 hours; proves the model is not silently a lie.
2. **T1.1** (real value factor) — 1–2 evenings; single highest-impact fix.
3. **T2.1** (regime overlay) — 1 evening; drawdown protection before live orders resume.
4. **T2.2** (correlation clustering) — 1 evening; effective diversification.
5. **T1.2** (survivorship bias haircut) — 1 hour; honesty in reporting.
6. **T2.3, T3.** — as time permits.

Each closed item is logged in §**??** with the closing commit SHA.

3.5 Strategy at a Glance

Metric	Value
CAGR	**12.1%**
Sharpe Ratio	**0.64**
Sortino Ratio	**0.95**
Maximum Drawdown	-22.2%
Calmar Ratio	0.55
Annualised Volatility	12.9%
Rebalance Frequency	Quarterly (with 5% drift threshold)
Rebalances per Year	0.6
Number of Positions	30
Optimizer	Robust MVO (Ledoit-Wolf + Michaud resampling)
ETF Universe	999 liquid (from 4,953 total, 2,184 non-leveraged)
Backtest Period	February 2021 – February 2026 (5 years)
Data Source	Interactive Brokers historical API

Table 1: Key Strategy Metrics

3.6 SPY Benchmark Comparison

Metric	Portfolio	SPY	Difference
CAGR	12.1%	13.4%	-1.3pp
Sharpe Ratio	**0.64**	0.59	**+0.05**
Volatility	**12.9%**	17.1%	**-4.2pp**
Max Drawdown	**-22.2%**	-24.5%	**+2.3pp**
Final Value ($1M start)	$1,764,103	$1,843,000	—

Table 2: Portfolio vs SPY Benchmark (Feb 2021 – Feb 2026)

The strategy trades 1.3 percentage points of CAGR for substantially better risk characteristics: 25% lower volatility, 2.3pp shallower maximum drawdown, and a higher Sharpe ratio. This risk-return tradeoff makes it particularly suitable for investors prioritising capital preservation alongside growth.

3.7 Performance Summary

Version 4.2 represents a major upgrade from Version 3.0:

- **3.4× larger universe**: 4,953 ETFs (up from 1,459) with 999 liquid candidates
- **Robust MVO optimizer**: Upgraded from RankBased to robust Mean-Variance Optimisation with Ledoit-Wolf covariance shrinkage, Bayes-Stein return shrinkage, and Michaud resampling (see Section 2.2)
- **Better diversification**: 30 positions (up from 20) with 3–8% weight bounds

- **Lower rebalancing**: Quarterly with 0.6 rebalances/year
- **International diversification**: Natural tilt toward international dividend ETFs reduces portfolio volatility 25% below SPY
- **Robust across regimes**: Validated through COVID recovery, 2022 bear market, and 2023–25 rally

3.8 Capital Structure

Component	Amount	Purpose
Total Portfolio	$1,000,000	Backtest reference capital
Active Positions	$960,000	30 positions at ≈$32,000 each
Cash Reserve	$40,000	Maintained for rebalancing

Table 3: Capital Allocation Structure

4 Changes from Version 3.0 and Why They Matter

4.1 Universe Expansion: 1,459 → 4,953 ETFs

Attribute	v3.0	v4.0
Raw Tickers Collected	3,214	**4,953**
After Quality Filter (≥252 days)	1,459	2,184
After Leveraged/Inverse Filter	1,459	2,184
After Liquidity Filter (≥50K avg vol)	Not applied	**999**

Table 4: Universe Evolution

Why this matters: The IB data collection is now essentially complete (4,953 of ∼5,000 US-listed ETFs). The key improvement in v4.0 is the addition of a **minimum daily volume filter** at 50,000 shares. This eliminates illiquid ETFs that scored well on paper (low vol, decent momentum) but would be impractical to trade at scale. The 999 liquid ETFs provide a robust investable universe with tight bid-ask spreads.

Volume filter impact on performance:

Volume Filter	N ETFs	CAGR	Sharpe	Max DD
No filter	2,184	11.7%	0.62	-21.8%
≥50K	**999**	**12.1%**	**0.64**	**-22.2%**
≥100K	731	11.1%	0.54	-24.7%

Table 5: Impact of Volume Filtering (MVO robust, 30 positions, quarterly)

The 50K filter is the sweet spot: it removes 1,185 illiquid ETFs while retaining 999 tradeable candidates. Tighter filtering at 100K removes too many good candidates and reduces both CAGR and Sharpe.

4.2 Optimizer: RankBased → Robust Mean-Variance Optimisation

Optimizer	Positions	CAGR	Sharpe	Max DD
RankBased (v3.0)	20	11.4%	0.58	-22.8%
RankBased	30	11.6%	0.61	-21.9%
MVO robust (v4.2)	**30**	**12.1%**	**0.64**	**-22.2%**

Table 6: Optimizer Comparison (50K+ volume, quarterly)

The problem with naïve MVO: Classical Mean-Variance Optimisation is well-documented as an "error maximiser" (Michaud, 1989). Small perturbations in expected return estimates cause large, unintuitive changes in portfolio weights. Axioma Research demonstrated that naïve MVO portfolios are dominated by estimation error: the optimizer allocates heavily to assets where expected returns are *most overestimated*, not where they are truly highest.

We confirmed this directly: perturbing expected returns by ±50 basis points and re-running naïve MVO with wide weight bounds (1–15%) produced **0% ticker stability**—completely different portfolios from the same data with trivial noise.

Why robust MVO is different: Our implementation addresses *every* known MVO failure mode through four layered defences:

MVO Criticism	Our Defence	Reference
Noisy covariance matrix	Ledoit-Wolf shrinkage	Ledoit & Wolf (2004)
Noisy expected returns	Bayes-Stein shrinkage (50%)	Jorion (1986)
Solution instability	Michaud resampling ($N = 50$)	Michaud (1998)
Extreme concentrations	Tight weight bounds (3–8%)	Standard practice
High-vol overweighting	Axioma risk penalty (γ)	Axioma Research

Table 7: How robust MVO addresses known failure modes

The tight weight bounds are critical: with 30 positions between 3% and 8% (equal weight = 3.3%), the optimizer can only make *marginal adjustments* around near-equal-weight. It cannot make wild bets. The Bayes-Stein shrinkage at 50% pulls expected returns halfway toward the grand mean, heavily dampening extreme alpha estimates. Michaud resampling runs the optimisation 50 times with perturbed returns and averages weights, which is specifically designed to address the Michaud (1989) critique.

This is how institutional allocators (BlackRock, AQR, etc.) actually deploy MVO in practice—not the textbook naïve version.

Why robust MVO outperforms RankBased: RankBased ignores correlations entirely, which means it cannot exploit diversification opportunities. The robust MVO considers the covariance matrix and identifies positions that are individually strong *and* provide hedging benefit to each other. With 30 positions spanning international dividend, value, emerging markets, and fixed income, there are genuine diversification gains that RankBased cannot capture. The result: +0.5pp CAGR, +0.03 Sharpe, and +0.06 Sortino.

RankBased remains available as a simpler alternative (`OPTIMIZER_TYPE = "rankbased"`) for users who prefer maximum weight transparency and deterministic outputs.

4.3 Position Count: 20 → 30

Increasing from 20 to 30 positions:

- **Lower concentration risk**: HHI decreases from 0.055 to 0.040
- **Better Sharpe ratio**: 0.64 vs 0.59 (30 vs 20 positions with MVO)

- **Lower max drawdown**: -22.2% vs -25.7%
- **CAGR improvement**: 12.1% vs 10.5% (30 positions improves CAGR with MVO due to diversification benefit)

With 999 eligible ETFs and the geometric-mean factor model, the top 30 remain high-quality candidates. Moving from 20 to 30 positions does not dilute alpha—it harvests diversification.

4.4 Rebalancing: Bimonthly → Quarterly

Frequency	CAGR	Sharpe	Reb/Year	Max DD
Monthly	11.7%	0.62	0.6	-21.8%
Bimonthly	11.7%	0.62	0.6	-21.8%
Quarterly	**12.1%**	**0.64**	**0.6**	**-22.2%**

Table 8: Rebalance Frequency Comparison (MVO robust, 30 positions, 50K+ vol)

Quarterly rebalancing marginally outperforms bimonthly and monthly. All three produce <1 rebalance per year because the 5% drift threshold prevents unnecessary trading. Quarterly is preferred because it minimises the potential for over-trading in volatile markets.

4.5 Quantified Impact: v3.0 vs v4.0

Metric	v3.0	v4.0	Change
CAGR	12.3%	12.1%	-0.2pp
Sharpe Ratio	0.66	0.64	-0.02
Sortino Ratio	0.95	0.95	unchanged
Max Drawdown	-23.4%	-22.2%	**+1.2pp better**
Calmar Ratio	—	0.55	(new metric)
Volatility	—	12.9%	(new metric)
ETF Universe	1,459	999 (liquid)	See note
Positions	20	**30**	+50%
Optimizer	RankBased	**Robust MVO**	Upgrade (see Section 2.2)
HHI (concentration)	∼0.055	**0.040**	-27%
Rebalances/Year	<1	0.6	Unchanged
Transaction Costs	—	$755	$151/yr

Table 9: Version 3.0 vs 4.0 Comparison

Note: Although the headline CAGR is marginally lower (12.1% vs 12.3%), the v4.2 portfolio is substantially better risk-adjusted. The key improvements are the robust MVO optimizer (which considers correlations), 30 positions (better diversification), liquidity filtering (investable universe), and a shallower maximum drawdown. The Sortino ratio is unchanged at 0.95, confirming equivalent downside-adjusted performance.

5 Why the Expanded Universe Improves the Strategy

The expansion from ~1,500 to ~5,000 ETFs—and the subsequent filtering to 999 liquid candidates—improves the strategy in four distinct ways:

5.1 Better Factor Concentration

With 999 ETFs scored on four factors versus the previous 1,459 (many illiquid), the top 30 represent a more selective percentile cut. The geometric-mean integration penalises ETFs with weakness on any factor, so a larger pool yields ETFs that are genuinely strong across *all four* dimensions. The top-30 integrated score threshold rises from ~0.78 (v3.0) to ~0.82 (v4.0), indicating better factor quality.

5.2 International Diversification

The factor model naturally selects international dividend and value ETFs because they score well on:

- **Quality**: Higher Sharpe ratios from lower volatility
- **Low Volatility**: International developed markets have lower vol than US tech-heavy indices
- **Value**: Low expense ratios (broad international ETFs charge 0.05–0.30%)
- **Momentum**: International markets showed strong momentum in 2024–2026

This international tilt is not a weakness—it is the primary driver of the portfolio's 25% lower volatility versus SPY. Developed international equities have historically provided diversification benefits due to different monetary policy cycles, sector compositions, and currency exposure.

5.3 Lower Correlation Across Positions

With 999 liquid candidates spanning US equity, international developed, emerging markets, sectors, and fixed income, the robust MVO optimizer explicitly considers pairwise covariance when constructing the 30-position portfolio. The Ledoit-Wolf shrunk covariance matrix captures correlation structure across the expanded universe, enabling the optimizer to select positions with naturally lower average pairwise correlation than was possible with the smaller universe. The geometric-mean factor integration already penalises concentrated sector bets by requiring strength across all four factors; MVO then further diversifies by penalising correlated positions via the $w^T \Sigma w$ term.

5.4 Practical Investability

The 50K+ volume filter ensures all selected ETFs are practically tradeable:

- Average daily volume of selected positions: 50K – 27M shares
- Tight bid-ask spreads: typically <0.05% for these volumes
- A $1M portfolio with 30 positions (~$33K each) represents <0.1% of daily volume for all positions
- No market impact concerns at this portfolio scale

6 Portfolio Composition and Diversification

6.1 Current Holdings (as of February 2026)

#	Ticker	Weight	Avg Vol	Category
1	EFV	8.8%	2,617K	Intl Value
2	VYMI	8.4%	454K	Intl Dividend
3	EWP	4.4%	523K	Spain
4	AVDV	3.6%	326K	Intl Multi-Factor
5	IVLU	3.5%	470K	Intl Value
6	FDD	3.4%	137K	Intl Dividend
7	PXF	3.4%	104K	Intl Fundamental
8	IDV	3.2%	790K	Intl Dividend
9	IQDF	3.2%	70K	Intl Quality Dividend
10	FGD	3.2%	127K	Intl Dividend
11–20	*RODM, VEA, DFAX, AVEM, HFXI, FNDC, GCOW, VEU, FNDE, VXUS*			
21–30	*ACWX, HEFA, QEFA, SCHY, EEM, IGF, SDIV, FEMB, RWX, LEMB*			

Table 10: Portfolio Holdings (30 positions)

6.2 Concentration Analysis

Metric	Value
Number of Positions	30
Maximum Weight	8.8% (EFV)
Minimum Weight	2.0% (LEMB)
Mean Weight	3.3%
HHI (Herfindahl)	0.040
Top 5 Concentration	28.7%
Top 10 Concentration	46.1%

Table 11: Portfolio Concentration Metrics

The HHI of 0.040 indicates a well-diversified portfolio. For reference, an equal-weight 30-position portfolio would have HHI = 0.033. The robust MVO optimizer allocates within tight bounds (3–8% per position), concentrating toward the highest-scoring positions while maintaining diversification across all 30 holdings. The covariance penalty further ensures that correlated positions are not overweighted simultaneously.

6.3 Geographic and Category Allocation

Category	Weight
International Dividend	14.4%
International Broad Developed	11.7%
Multi-Factor	6.6%
International Single Country	4.4%
International Developed SmallCap	2.9%
Emerging Broad	2.9%
Emerging Asia	2.7%
Infrastructure	2.6%
International Bonds	2.0%
Other/Uncategorized	49.7%

Table 12: Category Allocation (the "Uncategorized" bucket contains mostly international equity and multi-asset ETFs not yet in the curated category list)

The portfolio's international tilt is a deliberate outcome of the factor model. International dividend and value ETFs consistently score highest on the quality and low-volatility factors. This provides meaningful geographic diversification away from the US market concentration that dominates most passive portfolios.

6.4 Rebalancing Frequency

Date	Positions	Portfolio Value
2021-02-16	30	$1,000,000 (initial)
2022-09-21	30	$893,154
2022-10-03	30	$859,088

Table 13: Rebalancing History (only 3 rebalances in 5 years)

The strategy rebalanced only **3 times in 5 years** (0.6 times per year). The first was the initial portfolio construction. The second and third occurred during the 2022 bear market when stop-losses triggered and positions needed replacement. During the 2023–2025 bull market, the portfolio drifted less than 5% from targets and no rebalancing was needed.

This ultra-low turnover produces:

- **Total transaction costs**: $755 over 5 years ($151/year)
- **Tax efficiency**: Fewer than 1 taxable event per year
- **Minimal market impact**: Only 117 trades over 5 years

7 Factor Framework

7.1 Factor Definitions

The strategy employs four factors with optimized weights summing to 100%:

7.1.1 Momentum Factor (35%)

Calculation:

$$\text{Momentum}_i = \frac{P_{i,t-21} - P_{i,t-252}}{P_{i,t-252}} \tag{1}$$

Where:

- $P_{i,t-21}$ = price 21 trading days ago (skip recent month)
- $P_{i,t-252}$ = price 252 trading days ago ($\approx$ 1 year)
- Skipping the most recent 21 days avoids short-term reversal effects
- Winsorize at 1st/99th percentile to limit outlier influence

Purpose: Capture trending ETFs with strong 12-month performance while avoiding the last-month reversal documented in the academic literature.

Academic Basis: Jegadeesh & Titman (1993) momentum anomaly; the 1-month skip follows Novy-Marx (2012).

7.1.2 Quality Factor (30%)

Components:

$$\text{Sharpe Ratio} = \frac{\mu_r - r_f}{\sigma_r}\sqrt{252} \quad \text{(40\% weight)} \tag{2}$$

$$\text{Drawdown Resilience} = -1 \times \text{MaxDD} \quad \text{(30\% weight)} \tag{3}$$

$$\text{Return Stability} = -1 \times \sigma_r\sqrt{252} \quad \text{(30\% weight)} \tag{4}$$

Each component is z-score normalised and combined with the listed weights.

Purpose: Select ETFs with consistent, high risk-adjusted returns.

Academic Basis: Asness, Frazzini, Pedersen (2019) quality investing.

7.1.3 Volatility Factor (20%)

Calculation:

$$\text{Volatility}_i = \frac{1}{\sigma_{60d,i} \times \sqrt{252}} \tag{5}$$

Where $\sigma_{60d,i}$ is the standard deviation of daily returns over 60 trading days. The inverse ensures low-volatility ETFs score higher.

Purpose: Tilt toward stable ETFs to reduce portfolio-level volatility.

Academic Basis: Baker, Bradley, Wurgler (2011) low volatility anomaly.

7.1.4 Value Factor (15%)

Calculation:

$$\text{Value}_i = -1 \times \text{Expense Ratio}_i \tag{6}$$

Lower expense ratios represent better value. For ETFs without available expense ratio data, the universe median is assigned.

Purpose: Prefer lower-cost ETFs, reducing permanent drag on returns.

7.2 Factor Integration

Factors are combined using a **weighted geometric mean**:

$$\text{Score}_i = \text{Mom}_i^{0.35} \times \text{Qual}_i^{0.30} \times \text{Vol}_i^{0.20} \times \text{Val}_i^{0.15} \tag{7}$$

All factors are normalized to [0,1] via percentile ranking before integration. ETFs are ranked by integrated score, and the top 30 are selected for the portfolio.

Why geometric mean: The geometric mean penalises ETFs with weakness on any single factor. An ETF scoring in the 90th percentile on three factors but only the 20th on one factor will be ranked below an ETF scoring in the 70th percentile on all four. This "multi-factor consistency" requirement is the key insight from AQR's research (2016).

7.3 Justification of the Weight Vector 35/30/20/15

The weights momentum = 35%, quality = 30%, volatility = 20%, value = 15% are an *informed prior*, not an in-sample optimum. This subsection records the reasoning explicitly following the July 2026 recalibration review (`docs/research/2026-07_factor_weight_diagnostic.md`).

7.3.1 The empirical diagnostic and why we did not act on it

A level-1 grid search over the three-factor simplex (momentum, quality, volatility; value excluded because it is proxied only by expense ratio and carries no meaningful time-series signal) was run on the cached ETF universe from 2021-04-12 to 2026-04-16, monthly rebalance, top-20 by integrated rank, exponential weights, 5 bps turnover cost. The in-sample winners across both a 5-year and a 3-year window were uniformly *momentum-heavy*: 80–90% momentum, 0–30% quality, 10–20% low-volatility. Sharpe ratios reached 1.30 on the 3-year slice compared with $\approx$0.96 for the current 35/30/20/15 policy.

We deliberately declined to update the policy on this evidence. The 2021–2026 window was overwhelmingly a single momentum-friendly regime (post-COVID rally, mega-cap technology concentration, AI narrative from 2023 onward). Both the full and 3-year windows overlap and sample the same regime; this is not an independent invariance test. Adopting the mechanical in-sample winner (90% momentum, 0% quality, 10% low-vol) would be an act of confirmation bias, not calibration — exactly the exposure the operator is on record as most concerned about (a potential unwinding of the AI/tech valuation regime).

7.3.2 The ergodicity argument for insurance-premium factors

The stronger justification for retaining quality and low-volatility weights is that equity returns are *multiplicative and non-ergodic.* Following Peters (2019) and the ergodicity-economics literature, for a single investor on a single path the arithmetic mean of returns diverges from the time-average growth rate that they actually experience. By Jensen's inequality, since $\log(\cdot)$ is concave:

$$\underbrace{\mathbb{E}[\log(1+r)]}_{\text{time-average growth rate}} \quad < \quad \underbrace{\log(1+\mathbb{E}[r])}_{\text{arithmetic-mean growth}} \tag{8}$$

The gap — the "ergodicity premium" — is lost whenever the strategy is optimised against arithmetic expected return. This has two direct consequences for factor-weight selection:

1. **Drawdown asymmetry.** A 50% drawdown requires a 100% subsequent gain to break even, not a 50% one. Arithmetic-mean Sharpe optimisation understates this. Factors that reduce drawdown variance (quality, low-volatility) buy protection against exactly the multiplicative losses that a single-path investor most needs to avoid.

2. **Long-window Capital Market Assumption critique.** A 50-year CMA of "US equities return 7% real" presumes ergodicity — that averaging over many parallel investors or a long horizon converges to a well-defined distribution. It does not. Any given investor experiences one path, and the arithmetic prior overstates what they will earn. Long-window calibration exercises therefore *cannot* be treated as sufficient basis for weight-vector selection in a live-money system; the ensemble picture is not the individual-path picture.

The 30% quality and 20% low-volatility allocations in the current policy are, under this reading, **insurance premia against non-ergodic tail losses, not underperforming factors**. In-sample they read as drag on Sharpe. On the actual path the investor walks they may be the reason the strategy survives to compound. This is precisely the Kelly / log-utility argument (Kelly, 1956; Thorp, 2006) applied to factor construction.

7.3.3 The Bayesian-prior critique

A parallel objection to long-window optimisation is that Bayesian methods with human-elicited priors are systematically prone to salience, recency, and narrative bias. A prior distribution shaped by a decade of momentum-factor success will "know" momentum works; that is not knowledge, it is regime-embedded belief. This is a second reason to refuse to let a five-year in-sample winner override an informed prior — the numerical evidence and the elicited prior are contaminated by the same regime.

7.3.4 Why these specific integer values

The specific magnitudes 35/30/20/15 are not derived from a proven optimum. They are chosen to satisfy three constraints:

1. **Momentum > quality > volatility > value.** Reflects the ordering of empirical premia in the AQR literature (Jegadeesh & Titman, 1993; Asness, Frazzini & Pedersen, 2019; Baker, Bradley & Wurgler, 2011) and the compounded issue that the value factor here is proxied only by expense ratio.
2. **No factor at or below 10%.** A factor below 10% receives such a small weight in the weighted geometric mean that it fails to discriminate between candidates; the multi-factor consistency requirement collapses to a three-factor screen.
3. **No factor above 40%.** A factor above 40% dominates the integrated score to the point that a single-factor selection would produce nearly the same portfolio. This defeats the multi-factor rationale and re-introduces single-factor tail risk.

Within those constraints, the round-number allocation is a Schelling point that can be explained in a sentence to a future maintainer or auditor. It is deliberately regularised away from an in-sample optimum.

7.3.5 When to revise

The policy weights will be revisited when at least one of the following is available:

1. **Backfilled history through pre-2020** that permits a non-overlapping regime comparison (2011–2015 vs 2021–2026 is the immediate target).
2. **A working value factor** using fundamental data (P/E, P/B, dividend yield) rather than expense ratio.
3. **A deliberate factor-level sentiment override** — a documented, time-stamped, committed edit to `configs/etf_smart_beta.toml` reflecting a macro view (e.g. a downweighting of momentum during a period the operator judges to be a valuation-driven late-cycle

regime). Such overrides are the operator's discretionary layer above the systematic core, and are logged in git rather than automated.

References for this subsection:

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8 Portfolio Construction

8.1 Universe Filtering Pipeline

Starting from 4,953 ETFs with IB historical data:

1. **History requirement:** $\geq$252 trading days of price data $\rightarrow$ 2,254 ETFs
2. **Data quality:** <20% missing daily bars $\rightarrow$ 2,254 ETFs
3. **Leveraged/Inverse exclusion:** Remove 70 leveraged/inverse ETFs $\rightarrow$ 2,184 ETFs
4. **Liquidity filter:** $\geq$50,000 average daily volume $\rightarrow$ **999 ETFs**

8.2 Robust Mean-Variance Optimisation (Default)

The strategy uses robust MVO as the default optimizer (`OPTIMIZER_TYPE = "mvo"`). The implementation uses an Axioma-style risk adjustment with three layers of robustness:

$$\max_{w} \quad \mu^T w - \lambda\, w^T \Sigma\, w - \gamma\, w^T \sigma \tag{9}$$

Subject to: $\sum_i w_i = 1, \quad 0.03 \leq w_i \leq 0.08$

Robustness layers:

1. **Ledoit-Wolf covariance shrinkage** (Ledoit & Wolf, 2004): Shrinks the sample covariance toward a structured target, reducing estimation noise in the Σ matrix
2. **Bayes-Stein return shrinkage** (Jorion, 1986): Shrinks expected returns toward the cross-sectional mean at 50% strength, dampening extreme alpha estimates that cause naive MVO to fail
3. **Michaud resampling** (Michaud, 1998): Runs MVO 50 times with perturbed returns and averages the resulting weights, directly addressing the "optimization enigma" (Michaud, 1989)
4. **Tight weight bounds** (3–8%): With 30 positions summing to 100% (equal weight = 3.33%), the optimizer can only make marginal adjustments from equal weight—it cannot make wild bets
5. **Axioma risk penalty** ($\gamma \cdot w^T \sigma$): Explicitly penalises high-volatility positions, preventing the optimizer from concentrating in volatile assets

Advantages over naive MVO:

- **Exploits correlation structure**: The $w^T \Sigma w$ term actively diversifies across correlated positions—something rank-based methods cannot do
- **Stability**: Tight bounds + shrinkage + resampling produce stable weights across runs. Each position can only vary $\pm$5pp from equal weight
- **Institutional standard**: This is how BlackRock, AQR, and other quantitative allocators deploy MVO in practice (Axioma, 2013)

8.3 RankBased Exponential Weighting (Alternative)

RankBased weighting is available as a configurable option (`OPTIMIZER_TYPE = "rankbased"`). After the top N ETFs are selected by integrated factor score, weights are assigned by rank:

$$w_i = \frac{e^{-\alpha \cdot r_i}}{\sum_{j=1}^{N} e^{-\alpha \cdot r_j}}, \quad r_i = \text{rank of ETF } i \text{ (0-indexed)} \tag{10}$$

Where α controls the steepness of the exponential decay. This is a stable, transparent alternative that ignores correlation structure entirely. It may be preferred when the user values deterministic weights (identical inputs always produce identical outputs) over risk-adjusted optimality.

Tradeoff: RankBased does not consider correlations between positions. Two highly correlated ETFs with similar factor scores will both receive high weights, reducing effective diversification. Robust MVO penalises this via the covariance matrix.

8.4 Rebalancing Rules

Frequency: Quarterly

Drift Threshold: 5% — a scheduled rebalance is skipped if no position has drifted more than 5% from target weight. This results in 0.6 actual rebalances per year.

Rebalance Actions:

- Sell positions no longer in the top 30
- Buy new positions that have entered the top 30
- Adjust existing positions that have drifted >5% from target weight
- Maintain cash reserve for rebalancing

9 Risk Management Framework

9.1 Automated Trailing Stops

Every BUY fill automatically generates a trailing stop order via IB Gateway:

Parameter	Value
Order Type	TRAIL
Trail Amount	10%
Time in Force	GTC (Good-Til-Cancelled)
Trigger	LAST price
Outside RTH	No

Table 14: Trailing Stop Configuration

9.2 Entry Stop-Loss

The backtesting engine applies a 12% entry-based stop-loss:

$$\text{If } P_{i,t} < P_{i,\text{entry}} \times 0.88 \text{ then SELL} \tag{11}$$

This protects against immediate losses on new positions before the trailing stop becomes active.

9.3 Position-Level Risk Controls

Control	Limit
Maximum Loss Per Position	-12% (entry stop)
Trailing Protection	10% from peak (TRAIL order)
Position Concentration	3–8% per position (MVO bounds)
Maximum Weight	8% (hard constraint)
Number of Positions	30
Leveraged/Inverse ETFs	Excluded from universe

Table 15: Position-Level Risk Limits

10 Extended Backtest Results

10.1 Configuration Comparison

Full results across all tested configurations (50K+ volume filter):

Optimizer	Pos	Freq	CAGR	Sharpe	Sortino	Max DD	Reb/Yr
RankBased	20	Bimth	11.4%	0.58	0.85	-22.8%	1.0
RankBased	25	Bimth	10.9%	0.55	0.81	-23.4%	1.0
RankBased	30	Bimth	11.6%	0.61	0.89	-21.9%	0.6
RankBased	30	Qtrly	11.6%	0.61	0.89	-21.9%	<1
MVO (robust)	20	Bimth	11.5%	0.59	0.87	-22.2%	1.0
MVO (robust)	25	Bimth	11.3%	0.59	0.86	-22.5%	0.6
MVO (robust)	30	Bimth	11.7%	0.62	0.90	-21.8%	0.6
MVO (robust)	**30**	**Qtrly**	**12.1%**	**0.64**	**0.95**	**-22.2%**	**0.6**

Table 16: Full Configuration Comparison (50K+ vol filter, 5-year backtest)

10.2 Universe Scope Comparison

Testing with different universe definitions (MVO robust, 30 positions):

Universe	N	CAGR	Sharpe	Max DD	Calmar	Reb/Yr
All liquid	999	**12.1%**	**0.64**	-22.2%	0.55	0.6
US-focused	900	11.0%	0.59	**-17.9%**	**0.61**	0.6
Equity only	907	8.7%	0.39	-25.8%	0.34	1.0

Table 17: Universe Scope Comparison

Observation: The US-focused universe (excluding international ETFs) produces the lowest maximum drawdown (-17.9%) and highest Calmar ratio (0.61), but at the cost of lower CAGR. The all-liquid universe provides the best risk-adjusted returns (highest Sharpe) by leveraging international diversification.

10.3 Historical Version Comparison

Version	Universe	CAGR	Sharpe	Sortino	Max DD
v1.0 (yfinance, 7mo)	623	9.6%	0.83*	—	-7.95%*
v2.0 (yfinance, 5yr)	623	9.1%	0.40	0.55	-27.2%
v3.0 (IB, 5yr)	1,459	12.3%	0.66	0.95	-23.4%
v4.2 (IB expanded, robust MVO)	999	12.1%	0.64	0.95	-22.2%

Table 18: Strategy Evolution Across Versions

**v1.0 metrics measured over a benign 7-month period and are not comparable to the 5-year backtests.*

11 Implementation: Automated Pipeline

11.1 Architecture Overview

The strategy is implemented as a 7-step automated pipeline. Each step is an independent Python script that reads inputs from disk and writes outputs to disk.

Step	Script	Purpose	Output
1	s1_universe.py	ETF universe discovery	eligible_tickers.txt
2	s2_collect.py	IB historical data collection	Per-ticker parquets
3	s3_factors.py	Factor scoring	factor_scores_latest.parquet
4	s4_optimize.py	Portfolio optimization (robust MVO)	target_portfolio.csv
5	s5_backtest.py	Backtesting & performance	backtest_results.csv
6	s6_trades.py	Trade recommendations	trade_plan.csv
7	s7_execute.py	IB order execution + stops	execution_log.csv

Table 19: Pipeline Steps

11.2 IB Gateway Integration

Connection: Port 4001 (IB Gateway), client ID 5
Capabilities:

- Historical data download (5 years, daily bars, rate-limited at 12s intervals)
- Live portfolio positions and account summary
- Market/limit order placement
- Automated TRAIL stop orders on all BUY fills
- Resume support for interrupted data collection (per-ticker parquet caching)

11.3 Data Collection Status

- **Total collected:** 4,953 ETFs (essentially complete)
- **History depth:** 5 years (1,256 trading days) for most ETFs
- **Liquid universe:** 999 ETFs with ≥50K average daily volume
- **All data cached:** Per-ticker parquet files, fully resumable

11.4 Trade Execution Flow

1. Connect to IB Gateway (port 4001)
2. Pull live positions and account summary
3. Compare current vs target portfolio (MVO optimal weights)
4. Generate trade plan: sell non-targets, buy missing, rebalance drifted (>5%)
5. Enforce cash reserve on all buys
6. Execute orders with `CONFIRM = True` safety gate
7. For every BUY fill: immediately place 10% TRAIL stop (GTC)
8. Log all trades to execution log

12 Configuration Reference

Parameter	**Value**
Initial Capital	\$1,000,000
Active Positions	30
Optimizer	Robust MVO (Ledoit-Wolf + Michaud resampling)
Risk Aversion (λ)	2.0
Axioma Risk Penalty (γ)	0.5
Min Weight	3% (hard constraint)
Max Weight	8% (hard constraint)
Return Shrinkage	50% Bayes-Stein toward cross-sectional mean
Resampling Iterations	50 (Michaud)
Entry Stop-Loss	12% from entry price
Trailing Stop	10% TRAIL, GTC (on all BUY fills)
Rebalance Frequency	Quarterly
Drift Threshold	5%
Commission	\$0 (IB US ETF trades)
Spread Cost	2 basis points
Slippage	2 basis points
Min Daily Volume	50,000 shares
Factor Weights	
Momentum	35% (252-day, skip 21)
Quality	30% (Sharpe + Resilience + Stability)
Volatility	20% (inverse 60-day vol)
Value	15% (inverse expense ratio)

Table 20: Full Strategy Configuration

13 Limitations and Risks

1. **Static factor scores:** The current backtest uses end-of-period factor scores applied retrospectively. A fully dynamic (rolling) backtest would provide more realistic results, potentially with slightly lower CAGR due to lag.

2. **International bias:** The factor model naturally selects international ETFs. If international markets underperform the US for an extended period, the strategy will lag SPY. This is the tradeoff for lower volatility.

3. **No sector constraints:** The optimizer does not enforce sector diversification beyond what the factor model naturally provides. Adding explicit sector caps (e.g., max 30% per category) is a planned enhancement.

4. **Survivorship bias:** The universe includes only ETFs currently listed. ETFs that were delisted during the backtest period are not included, which may slightly overstate performance.

5. **Expense ratio data:** Real expense ratios are not yet reliably available for the full universe. When expense data is missing, the Value factor (15% weight) is automatically skipped and its weight redistributed proportionally across the remaining three factors. Incorporating real expense data would restore the Value factor signal.

6. **MVO estimation risk:** Although the robust MVO implementation addresses the classical failure modes identified by Michaud (1989)—via Ledoit-Wolf covariance shrinkage, Bayes-Stein return shrinkage, and Michaud resampling—the optimizer still relies on estimated expected returns and covariance. Tight weight bounds (3–8%) are the primary safeguard: with 30 positions at equal weight 3.33%, the optimizer can only deviate ±5pp from equal weight, limiting the damage from estimation error. Users should monitor weight stability across consecutive runs.

14 Next Steps

1. **Rolling factor model:** Implement dynamic factor recalculation at each rebalance date (rather than static end-of-period scores) for more realistic backtest validation.

2. **Real expense ratios:** Integrate actual ETF expense ratio data to improve the Value factor signal.

3. **Sector constraints:** Add explicit category concentration limits (max 30% per sector) to the portfolio optimizer.

4. **Live deployment:** Execute the pipeline against real IB positions. All infrastructure is in place; requires setting `CONFIRM = True` in the execution notebook.

5. **Performance monitoring:** Track live performance against backtest expectations. Update this document with live results at the first quarterly review.

15 Academic References

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16 Change Log

Entries are chronological, most recent first. Each entry cites the git commit that closes it (short SHA). **Update rule:** every commit that touches `configs/etf_smart_beta.toml`, `src/factors/`, `src/portfolio/`, or the execution scripts must also append an entry here or explain why not.

2026-07-09 — Living Edition; adversarial review + weight justification

Commits: `d9826dc`, plus this one.

- Reframed the document as a *living operating document* rather than a fixed v4.2 spec. New §?? explains the reader contract, sources of truth, and update triggers.
- Added §?? pulling the live policy values from `configs/etf_smart_beta.toml` into a single source-of-truth table. Explicit acknowledgement that the body sections describing v4.x design details drift from the live policy in three places (position count 20 vs 30, weight bounds 2–15% vs 3–8%, default optimiser RankBased vs MVO).

- Added §?? with tiered flaws from the operator's own commissioned adversarial review:
  - T1.1 The "value" factor is fake (expense-ratio proxy only). Highest single-value fix.
  - T1.2 Survivorship bias in cached universe; mitigation via CAGR haircut.
  - T1.3 Look-ahead bias presumed absent but never proved by a pinning test.
  - T2.1 No regime awareness; proposed SPY-200d + VIX overlay.
  - T2.2 Correlation blindness in top-N selection; proposed clustering step.
  - T2.3 Conviction magnitude discarded.
  - T3.1–T3.3 Untested lookbacks, round-number risk params, unmeasured rebalance frequency.
- Added §?? (in the Factor Framework section) recording the July 2026 recalibration review. Records the ergodicity-economics case for retaining 35/30/20/15 as an insurance-premium-bearing informed prior rather than adopting the momentum-heavy in-sample winner. Cites Peters (2019), Kelly (1956), Thorp (2006), Taleb (2018).

2026-02-15 — Version 4.2 (previous major revision)

- Documented the v4.x design: 4,953-ETF universe, 30 positions, tight 3–8% weight bounds, robust MVO default with Ledoit-Wolf + Bayes-Stein + Michaud resampling, quarterly rebalance. This design was backtested and documented; the live policy has since diverged in the three places noted above.
- Recorded backtest results: 12.1% CAGR / 0.64 Sharpe / 0.95 Sortino / -22.2% MaxDD over Feb 2021–Feb 2026 for the v4.2 design.

Document status: Living
Last major review: 2026-07-09
Authoring: Operator with AI assistance; every commit signed via git